VA DIVISION A • 3RD GRADE • 2025–2026

Tower Building

Competition Project Slides

- Official Rules
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- Competition Tips

Build a freestanding tower using only paper straws & masking tape that holds 300 grams for 20 seconds.

📋 Event at a Glance

2	45	30	1 m	300g	20 s
Students per team	Min total session	Straws provided	Masking tape	Load to hold	Hold time

GOAL: Build the TALLEST freestanding tower possible that supports a 300-gram mass for 20 seconds. Score = Height + Written Assessment (each worth 50 points).

Materials:	30 paper straws (~20cm each) + 1 meter of 2.54cm masking tape. Scissors at event. NOTHING else.
Surface:	All support parts rest on the flat surface. No sticky tape may touch the surface.
Load test:	Team places 300g on the block and removes their hand. Timer starts immediately.
Unused straws:	Supervisor records unused straws — affects tiebreaker. Use nearly all 30!

Top platform:	Tower MUST have a flat area at the top to place the 5×5×0.6cm plywood block.
Height:	Measured to nearest mm BEFORE the load is placed. Block height is NOT counted.
Timer ends:	When tower fails OR 20 seconds elapse. Recorded to 1/100 of a second.
Safety:	ANSI Z87+ impact glasses required. Bring your own — not provided.

Official Scoring Formula

Scoring

Final Score (FS) = Assessment Score (TAS) + Height Score (THS) • Maximum = 100 points

Term	Formula / Rule	Notes
Assessment Score (TAS)	(Your score ÷ Best assessment score) × 50	Max 50 pts
Height Score (THS)	(Your height in mm ÷ Tallest height among all teams) × 50	Max 50 pts
Tiebreaker 1	Time to structural failure — longer hold = wins tie	After primary score
Tiebreaker 2	Fewest unused straws — use them all!	2nd tiebreaker
Tiebreaker 3	Best raw assessment score	Final tiebreaker

★ *Example: Best height = 450mm. Your height = 375mm → THS = (375÷450)×50 = 41.67 pts. Best assessment = 9, you score 4 → TAS = (4÷9)×50 = 22.22 pts. Final Score = 63.89*

Official Materials Kit

Materials

Paper drinking straws

~20 cm each. May be cut to any length or left whole.

30 straws — *Structural members*

Masking tape (2.54 cm / 1" wide)

May be cut to any shape or size. Only connection material allowed.

1 meter — *Connections only*

300-gram test mass

Placed on top of the wood block during the test phase.

Provided — *Load testing*

Plywood block (5×5×0.6 cm)

Placed on top of tower. Its height is NOT added to the measured tower height.

1 block — *Platform top*

Scissors

Provided at the event for cutting straws and tape.

At event — *Shared tool*

BRING: ANSI Z87+ impact safety glasses (mandatory, not provided) • Writing utensils • Optional: ruler, protractor

🔧 Build Strategy — 45 Minutes

Strategy

1 Assign roles immediately

0–2 min

One student leads the build; one takes the written test. Both must know the plan before the clock starts.

2 Design before building

2–7 min

Sketch the tower. Decide: square or triangular base? How many tiers? Allocate straws: ~6 base, ~18 upper, ~4 diagonals, 2 reserve.

3 Build base — triangulate first

7–18 min

Build the bottom tier and add diagonal braces BEFORE going up. A square frame with no diagonals collapses sideways immediately.

4 Build in tiers — brace each level

18–32 min

Add height in ~15 cm stages. Brace every tier before building the next. Check vertical alignment after every tier.

5 Build the top platform

32–38 min

Tape 2–3 straws in a flat cross pattern at the top. Must be flat and stable enough to hold the 5×5cm plywood block.

6 Use all straws + final checks

38–45 min

Add final diagonal braces with remaining straws. Unused straws hurt tiebreaker. Check plumb and all joints before time is called.

⚡ Engineering Tips — What Wins

Tips

Triangulate every rectangle

Add at least one diagonal to every rectangular frame. One diagonal straw turns a collapsible rectangle into two rigid triangles. This is the single most important rule.

Short columns buckle less

Halving column height makes it ~4× more resistant to buckling. Keep unsupported straw lengths to 15 cm or less. Add mid-height bracing on tall sections.

Wrap tape, don't just stick it

Wrapping tape around a joint creates a compression band — far stronger than tape that just covers the surface. Overlap straws 2–3 cm before taping.

Wide base, narrow top

A pyramidal shape keeps the center of gravity low and the base wide — making the tower much harder to topple when the 300g mass is placed on top.

Check plumb constantly

A lean that looks small at 20 cm looks disqualifying at 50 cm. Use a straw or pencil held vertically to check alignment after every tier.

Both parts are equal

Assessment and height are each worth 50 points. Know tower types, stability, center of gravity, and famous towers. A perfect test score beats a taller but weaker tower.